K.D.S.H.M. Perera

B.Sc. Engineering (Honours)

99B, Colombo Road, Pokunuwita, Sri Lanka

+94 721 982 710 | | | | | heshanm@uom.lk

Profile

My primary research interest lies in bio-inspired robotics, with a particular focus on unconventional locomotive platforms capable of navigating complex, unstructured environments. My work spans the full development pipeline of a robotic system, including mechanical design, hardware prototyping, embedded systems, mathematical modeling, simulation, and dynamic control. I am further interested in scaling these principles through swarm frameworks, deploying decentralized collectives of bio-inspired robots to solve complex tasks in unpredictable real-world settings.

Education

2021 – 2025 **B.Sc. Engineering (Hons) in Mechanical Engineering**
Specialisation: Mechatronic Systems Engineering, *University of Moratuwa*, Sri Lanka
GPA: 3.79 / 4.00
Class Rank: 10/120
Minor: Mathematics

2017 – 2019 **GCE Advanced Level, Physical Science Stream**, *D.S. Senanayake College, Colombo 07*
Results: 2 A's & 1 B; Z-Score: 2.3409

Work Experience

Feb 2026 – Present **Temporary Lecturer**, *Department of Mechanical Engineering, University of Moratuwa*, Moratuwa, Sri Lanka

Aug 2025 – Jan 2026 **Temporary Instructor**, *Department of Mechanical Engineering, University of Moratuwa*, Moratuwa, Sri Lanka

Mar 2024 – May 2024 **Intern / Student Researcher**, *Shandong University of Science and Technology*, Shandong, China

Nov 2023 – Feb 2024 **Intern / Project Engineer**, *Dipped Product PLC*, Sri Lanka

Research Projects

Underwater Drone for Coral Observation — Contributed to the design and development of a shallow-sea underwater drone using a novel fuzzy-adaptive sliding-mode controller for robust trajectory tracking; developed a VR-based teleoperation interface for real-time control and visualization.

Brachiating Robot — Designed, simulated, and built a prototype brachiation robot to investigate dynamic locomotion and control across variable bar spacing and heights, using a simplified control strategy where only the reaching hand is actively guided toward the next target, achieving brachiation in a single swing.

Biomimetic Fish Robot (Ongoing) — Developing a soft tendon-driven biomimetic fish robot featuring a non-contact magnetic actuation interface.

Wearable Balance Assistive Device (Ongoing)	**Co-supervising** the design and development of a wearable mechatronic device to provide active balance assistance for individuals with postural stability impairments.

Academic Projects

Environment Monitoring Station ▶	Implemented real-time environmental monitoring with signal conditioning, processing, remote communication and data visualization (web/mobile).
App-Controlled Mobile Robot	Developed a differential-drive mobile robot with image-based object detection, path planning, and manual/autonomous operation via a mobile app.
Micro-ROS Based Home Automation	Led development of an IoT smart-home prototype integrating micro-ROS for autonomous temperature-based fan control and web-based lighting/fan management.
Smart Garbage Collector	Collaboratively engineered a mobile robot using image processing to autonomously navigate predefined routes and collect litter.
Modular Microconveyor	Designed a modular microconveyor using chevron electrothermal actuators and a serpentine beam ratchet for precise, incremental motion; validated with multiphysics simulation.

Industrial Projects

Dynamic Work Bar Transfer Mechanism	Designed a pneumatic system to halt and transfer work bars between leaching tanks.
Web-Based SCADA Visualization	Developed a full-stack web application for real-time SCADA dashboards and historical data visualization.
Hydraulic Lifting Platform	Researched and prototyped a hydraulic lift for wheelchair accessibility (Shandong University of Science and Technology).

Achievements

Best Paper Award	Robotics & Intelligent Systems Track — Moratuwa Engineering Research Conference (MERCon 2025), Moratuwa, Sri Lanka.
The Institution Best Project	Institution of Mechanical Engineers (IMechE) — University of Moratuwa, Sri Lanka (2025).
Dean's List	Semesters 1, 2, 5, 7

Research Publications

- (Co-first author) *WS Ray: Extending Human Perception with a Teleoperated Underwater Drone for Subaquatic Environments*. International Conference on Automation Science and Engineering (CASE 2026). **Accepted**.
- (First author & presenter) *Brachiation Robot Inspired by Gibbon Dynamics: Insight for Control Across Irregular Bar Spacing and Heights*. Moratuwa Engineering Research Conference (MERCon 2025), Moratuwa, Sri Lanka. **Best Paper Award, Robotics & Intelligent Systems**.DOI: 10.1109/MERCon67903.2025.11217001.
- (First author & presenter) *A Fuzzy-Adaptive Sliding-Mode Control Strategy for Optimizing Control of an Underwater Drone*. MERCon 2025, Moratuwa, Sri Lanka.DOI: 10.1109/MERCon67903.2025.11217001.
- (Co-first author) *Development of Marine Grade Acrylonitrile – Butadiene – Styrene (ABS) based*

Hull of an Underwater Drone. MERCon 2025, Moratuwa, Sri Lanka. DOI: 10.1109/MERCon67903.2025.11217039.

- (Co-first author) *CFD-Driven Hydrodynamic Optimisation of a Whale Shark–Inspired Underwater Drone for Shallow Reef Observation*. 12th International Conference on Heat Transfer and Fluid Flow (HTFF 2025). DOI:10.11159/htff25.243.

- (First author) *A Chevron-Actuated Microconveyor for Precise Wafer-Scale Microfactory Transportation*. Mechanical Engineering Research Symposium 2025 (MERS 2025), Department of Mechanical Engineering, University of Moratuwa. Accepted.

Skills

Programming	Python, C++, MATLAB & Simulink , LabVIEW
Robotics & Simulation	ROS2, micro-ROS, Gazebo, MuJoCo, Webots, OpenCV
Design & Analysis	SolidWorks, AutoCAD, Ansys, Adams, Fluidsim
Embedded Systems & Electronics	Atmel Studio, Proteus, easyEDA, Arduino & ESP32, Raspberry Pi
Web & Data	HTML/CSS, JavaScript, SQL, Flutter
Office & Graphics	LaTeX, Microsoft Word, PowerPoint, Excel, Adobe Illustrator, Photoshop

Languages

Sinhala	Native
English	Fluent (TOEFL iBT: 5.0/6.0)

Mentorship & Service

2026|Committee Member	Editorial Commitee, MERS 2026, University of Moratuwa
2026|Lecturer	ME4463: Finite State Machine (FSM)
2026|Lecturer	ME3813: Machine Design Project - Motor Selection & Initial Calculations
2026|Lecturer	ME3261: Mechatronic System Design Project
2026|Lecturer	ME4740: Advanced Topics in Mechatronic System Engineering – Modeling & Simulation Section
2026|Lecturer	ME4191: Robotics & Automation Systems – ROS 2 Workshop
2026|Team Mentor	Singapore Autonomous Underwater Vehicle Challenge (SAUVC 2026)
2025|Instructor	ME3261:Mechatronic System Design Project – Original ROS 2 Workshop
2025|Instructor	Mechanical & Mechatronic Engineering Courses, University of Moratuwa
2023|Committee Member	Logistics Committee, EXMO 2023, University of Moratuwa
2023|Committee Member	Organizing Committee, MechatronX Inter-School Robotics Competition
2019|Member	Danuma Yathra Community Service Project
School Level|Member	Science Quiz Team Organizer, D.S. Senanayake College

References

Prof. Ranjith Amarasinghe
Professor
Department of Mechanical Engineering
University of Moratuwa
Tel: +94 71 8517882
Email: ranama@uom.lk

Dr. Dumith Jayathilaka
Stream Coordinator
Mechatronic Systems Engineering
University of Moratuwa
Tel: +94 11 2650301 ext. 4522
Email: dumithj@uom.lk